Name: ____________________________________

7.1 Angles of polygons.

1. Find the sum of the interior angle measures of a convex 17-gon.

2. Find each interior angle and each exterior angle of a regular hexagon.

3. The interior angles of a convex polygon total 2520°. How many sides?

4. Five angles of a hexagon measure 133°, 105°, 128°, 119°, and 111°. Find the sixth.

7.2 Properties of parallelograms.

5. In ▱ABCD, $AB = 7x - 10$, $CD = 4x + 17$, and $m\angle A = 63°$. Find x, AB, and $m\angle B$.

6. The diagonals of ▱JKLM meet at N. $JN = 8x - 13$ and $NL = 5x + 11$. Find x and JL.

7. Three vertices of ▱WXYZ are $W(-2, -1)$, $X(1, 4)$, and $Z(3, -3)$. Find Y and verify with midpoints.

7.3 Proving parallelograms.

8. Quadrilateral DEFG has $\angle D \cong \angle F$ and $\angle E \cong \angle G$. What can you conclude? Name the condition.

9. One pair of opposite sides of a quadrilateral measures $6t + 7$ and $9t - 20$, and those sides are parallel. Find t so the figure must be a parallelogram.

10. Is 'one pair of parallel sides and the other pair congruent' enough? Explain.

7.4 Special parallelograms.

11. In rhombus PQRS the diagonals meet at T and m∠QPR = 32°. Find m∠QPS, m∠PTQ, and m∠PQR.

12. In rectangle ABCD the diagonals meet at E. AC = 8x − 6 and BD = 5x + 18. Find x and AE.

13. In square JKLM the diagonals meet at X and JX = 7. Find JL, m∠JXK, m∠KJL, and JK (exact).

14. Always, sometimes, never: a rectangle is a rhombus.

7.5 Trapezoids and kites.

15. In kite ABCD, $\overline{AB} \cong \overline{AD}$ and $\overline{CB} \cong \overline{CD}$, m∠A = 44°, m∠C = 102°. Find m∠B and m∠D.

16. Isosceles trapezoid legs measure 10x − 19 and 6x + 21. Find x and each leg.

17. A trapezoid has bases 57 and 35. Find the midsegment.

18. A trapezoid's midsegment is 31 and one base is 44. Find the other base.

19. The bases of a trapezoid measure 12x − 9 and 8x + 5, and the midsegment measures 48. Find x and both bases.

20. In trapezoid PQRS, $\overline{PQ} \parallel \overline{SR}$, m∠P = 71°, and the trapezoid is isosceles. Find the other three angles.